



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-024A FleetPulse Intake & Recovery - Gate Review

Protected competency diagnosis | no copy-ready project

Artifact	C3_024A_FLEETPULSE_REVIEW_PROTECTED_RC1
Status	PROTECTED - OPEN AFTER FROZEN GATE ATTEMPT
Teaching standard	M01 beginner-first teacher loop. Videos are visual companions; the book remains sufficient fallback.
C2B boundary	Bridge foundations may be assumed, but C3 engineering behavior is demonstrated, broken, repaired, transferred and validated.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

Course rule: preserve what arrived -> validate interpretation -> quarantine ambiguity -> publish durably -> commit state last -> prove replay safety. A green job is never sufficient evidence by itself.



Diagnose from evidence, not score

For each G1-G7, mark PASS only if evidence proves the behavior on this case. One required FAIL blocks Gate PASS. Never compute an average or percentage.

DIAGNOSIS ORDER

1) raw/source identity; 2) grain/window; 3) parse/schema/time decision; 4) incremental/change/state rule; 5) failure before commit; 6) replay result; 7) independent reconciliation/tests; 8) logs/runbook.

G1 - Source contract & landing

PASS signal: Inputs/windows/grains declared; raw + manifest/fingerprint preserved. If weak, reopen only the relevant C3-01 lesson, rebuild a tiny changed example, close material, and save a new repair artifact.

G2 - Defensive parsing & evolution

PASS signal: CSV/JSON/schema/time behavior explicit; unsafe ambiguity quarantined/blocked. If weak, reopen only the relevant C3-01 lesson, rebuild a tiny changed example, close material, and save a new repair artifact.

G3 - Incremental/CDC semantics

PASS signal: Stable key/cursor/event semantics; ties, updates/deletes handled deliberately. If weak, reopen only the relevant C3-01 lesson, rebuild a tiny changed example, close material, and save a new repair artifact.

G4 - State commit & replay safety

PASS signal: Durable publish/validate before state; injected failure + restart proven. If weak, reopen only the relevant C3-01 lesson, rebuild a tiny changed example, close material, and save a new repair artifact.

G5 - Interface reliability

PASS signal: Auth secret boundary if relevant; timeout/pagination/retry/rate-limit rules justified. If weak, reopen only the relevant C3-01 lesson, rebuild a tiny changed example, close material, and save a new repair artifact.

G6 - Testing/reconciliation

PASS signal: Independent controls + edge/failure test; source/output/state reconciled. If weak, reopen only the relevant C3-01 lesson, rebuild a tiny changed example, close material, and save a new repair artifact.

G7 - Operational handoff

PASS signal: Sanitized logs, runbook, recovery/backfill policy, interview defense. If weak, reopen only the relevant C3-01 lesson, rebuild a tiny changed example, close material, and save a new repair artifact.



Critical-failure check

- Raw evidence overwritten or unavailable without acknowledged blocker.
- State/checkpoint advances past data that was not durably validated/published.
- Unresolved schema/time/key ambiguity silently accepted.
- Replay/restart skips eligible data or causes unintended accepted duplicates.
- Real credential exposed or fabricated evidence submitted.
- Independent reconciliation knowingly fails or is omitted for a critical boundary.
- Protected Gate review opened before first attempt was frozen.

AFTER DIAGNOSIS

Targeted repair does not auto-unlock another full Gate. A changed micro-retest can prove the repaired competency. Only explicit assignment opens Gate B/C for a fresh full retest.

Preserve attempt_001 unchanged. New work goes to repair_001 or the assigned next Gate attempt.